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Examples for utilizing comparison lemma

Ex. $\dot{x} = -(1+x^2)x$ $x(0) = a$

$$V = x^2$$

$$\dot{V} = 2x\dot{x} = -2x^2 - 2x^4$$

$$\leq -2x^2$$

$$= -2V$$

$$\dot{V} \leq -2V$$

$$V(t) \leq e^{-2t} V(0)$$

$$x^2(t) \leq e^{-2t} \cdot a^2$$

$$\Rightarrow |x(t)| \leq e^{-t} |a|$$

Ex. $\dot{x} = f(t, x)$
 $= -(1+x^2)x + e^t$

$$V = x^2$$

$$\dot{V} = 2x\dot{x}$$

$$= -2x^2 - 2x^4 + 2xe^t$$

$$\leq -2V + 2\sqrt{V} e^t$$

... (difficult to solve)

Alternate:

$$V = |x(t)| = \sqrt{x^2(t)}$$

$$\frac{dV}{dt} = \frac{2x(t)\dot{x}(t)}{2\sqrt{x^2(t)}}$$

$$= \frac{x(t) [-(1+x^2)x + e^t]}{\sqrt{x^2}}$$

$$= \frac{-x^2(1+x^2)}{\sqrt{x^2}} + \frac{x(t)e^t}{\sqrt{x^2}}$$

$$\leq -|x| (1+x^2) + e^t$$

$$\leq -|x| + e^t = -V + e^t$$

$$\dot{V} \leq -V + e^t$$

$$V(t) \leq e^{-t} V(0) + \int_0^t e^{-(t-\tau)} e^\tau d\tau$$

$$\leq e^{-t} V(0) + e^t \Big|_0^t$$

$$\leq e^{-t} V(0) + e^t - 1$$

$$|x(t)| \leq e^{-t} |a| + e^t - 1$$

Ex.

$$\begin{aligned} \dot{x}_1 &= -x_1 + \frac{2x_1}{1+x_2^2} \\ \dot{x}_2 &= -x_2 + \frac{2x_1}{1+x_1^2} \end{aligned}$$

$$V = x_1^2 + x_2^2$$

$$\dot{V} = x_1 \dot{x}_1 + x_2 \dot{x}_2$$

$$= -x_1^2 + \frac{2x_1x_2}{1+x_2^2} - x_2^2 + \frac{2x_1x_2}{1+x_1^2}$$

$$= -V + \frac{2x_1x_2}{1+x_2^2} + \frac{2x_1x_2}{1+x_1^2}$$

$$(x_1 - x_2)^2 = x_1^2 + x_2^2 - 2x_1x_2 \geq 0$$

$$\Rightarrow 2x_1x_2 \leq x_1^2 + x_2^2$$

$$4x_1x_2 \leq 2(x_1^2 + x_2^2)$$

$$\dot{V} \leq -V + 4x_1x_2$$

$$\leq -V + 2V$$

$$\leq V$$

$$V(t) \leq e^t V(0)$$

$$x_1^2(t) + x_2^2(t) \leq e^t (x_1^2(0) + x_2^2(0))$$

If $w_3(\cdot)$ is only semi-definite?

$$\text{i.e., } w_3(x) \geq 0 \quad \forall x \neq 0$$

Thm: Suppose $w_1(x) \leq V(t, x) \leq w_2(x)$

$$\dot{V} = \frac{\partial V}{\partial t} + \frac{\partial V}{\partial x} f(t, x) \leq -w_3(x)$$

w_1, w_2 are positive definite

w_3 is positive-semi definite

$-w_1$ is radially unbounded

then, $w_3(x(t)) \rightarrow 0$ as $t \rightarrow \infty$

Ex.
$$\left. \begin{aligned} \dot{x}_1 &= -x_1 + x_2 \\ \dot{x}_2 &= -x_1 \end{aligned} \right\} \dot{x} = Ax \quad A = \begin{bmatrix} -1 & 1 \\ -1 & 0 \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} -1-\lambda & 1 \\ -1 & -\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = \lambda(1+\lambda)+1$$

$$\therefore \lambda = \frac{-1 \pm \sqrt{1-4}}{2}$$

$$V = \frac{1}{2} x_1^2 + \frac{1}{2} x_2^2 \quad \left| \quad \begin{aligned} V &= \frac{1}{2} x^T P x \\ P &= I \end{aligned} \right.$$

$$\dot{V} = x_1 \dot{x}_1 + x_2 \dot{x}_2$$

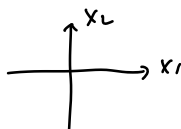
$$= x_1(-x_1 + x_2) + x_2(-x_1)$$

$$= -x_1^2$$

$$\dot{V} = 0 \quad | \quad S = \{x: x_1 = 0\}$$

$$M = \{0\}$$

$\therefore$ LaSalle $\Rightarrow x(t) \rightarrow 0$



Ex.

$$\begin{aligned}\dot{x}_1 &= -x_1 + \omega(t) x_2 \\ \dot{x}_2 &= -\omega(t) x_1 \\ \dot{x} &= A(t) x \\ &= \begin{bmatrix} -1 & \omega(t) \\ \omega(t) & 0 \end{bmatrix} x\end{aligned}$$

$$V = \frac{1}{2} x_1^2 + \frac{1}{2} x_2^2$$

$$\begin{aligned}\dot{V} &= x_1 \dot{x}_1 + x_2 \dot{x}_2 \\ &= x_1 (-x_1 + \omega(t) x_2) + x_2 (-\omega(t) x_1)\end{aligned}$$

$$= -x_1^2 \quad \cancel{- \left(\underbrace{f(x_1)}_{\text{P.D.}} + \underbrace{g(x_2)}_{\text{P.D.}} \right)} \quad \underline{\text{No}}$$

$x=0$ is uniformly stable

$$W_3(x) = x_1^2$$

$$x_1 \rightarrow 0 \text{ as } t \rightarrow \infty$$

$$\dot{x} = f(x, u(t))$$

$$\dot{x} = A(t) x$$

Thm: $x=0$ is uniformly asymptotically stable if and only if:

$$\|\Phi(t, t_0)\| \leq k e^{-\lambda(t-t_0)} \text{ for some } k > 0, \lambda > 0$$

$$x(t) = \Phi(t, t_0) x(t_0)$$

$$\dot{x} = Ax \Rightarrow x(t) = \underbrace{e^{A(t-t_0)}}_{\Phi(t, t_0)} x(t_0)$$

$$\equiv \Phi(t, t_0)$$

Recall

$$\Phi(t, 0) = \begin{bmatrix} e^{0.5t} \cos t & e^{-t} \sin t \\ e^{-0.5t} \sin t & e^{-t} \cos t \end{bmatrix}$$

$$\|\Phi(t, 0)\|_p^2 = e^{-2t} + e^t \cos^2 t + e^{-t} \sin^2 t$$

is not exponentially decaying

$\Rightarrow X=0$ is not uniformly stable